

Scalability & Future Plan

Date	04 july 2024
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Human Development Index (HDI) Prediction Using Machine Learning project can be enhanced to support more users, larger datasets, improved performance, and additional intelligent features. It also presents the future roadmap for expanding the application's capabilities.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Supports only a locally hosted	Limited accessibility for	High
2	Uses a Linear Regression model	Prediction accuracy can be	Medium
3	No user authentication or	Limited security and data	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a few local users	Deploy on cloud platforms such as
2	Data Storage	Dataset stored locally (CSV file)	Integrate a cloud database such as
3	Performance	Suitable for small datasets	Optimize code, caching, and use
4	Security	Basic local application	Implement user authentication,

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a	1 Month	Global accessibility and
Phase 2	Implement advanced	2 Months	Improved prediction
Phase 3	Add user login, prediction	3 Months	Better user experience and
Phase 4	Develop a mobile application and	4–6 Months	Cross-platform accessibility and

Conclusion

The Human Development Index (HDI) Prediction Using Machine Learning project has a strong foundation and can be scaled effectively to support larger datasets, more users, and additional intelligent features. By deploying the application to the cloud, improving the machine learning model, strengthening security, and expanding platform support, the project can evolve into a robust and production-ready solution.